

Who Chose Whom?

AP Statistics · Unit 1 · Topic 1.10 · TEACHER KEY

CED TETHER

- **VAR-1.E** Identify questions to be answered about data collection methods. [**Skill 1.A**]
- **VAR-1.E.1** Methods for data collection that do not rely on chance result in untrustworthy conclusions.

Essential question: How does the way people enter a sample limit which result we should trust?

DOK-3 skill: 4.A · **FRQ pattern:** voluntary-response-vs-random-which-number-and-why

DOK rationale: Part (c) coordinates selection method, response count, and target population to choose between competing estimates, explain why sample size cannot repair self-selection, and match a design change to the bias.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- In which method did students choose the sample, and in which did the researcher choose it?
- What kind of student is especially likely to click the website poll?
- What does a larger response count improve, and what can it not repair?
- How would you fix stop students from volunteering themselves into the sample?

[WARN] LOOK FOR

- Treating the larger response count as proof that the website sample represents the school.
- Calling the homeroom method an SRS of students; the random units are homerooms.
- Proposing a wording change that leaves the website poll open to self-selection.

SCORING PART (c) — E / P / I

Expected elements

- **chance-choice** (required) — Chooses 44 percent using chance selection rather than sample size
- **self-selection** (required) — Explains how website volunteers may differ and why 412 does not remove that problem
- **matched-fix** (required) — Proposes chance selection from the school population instead of open volunteering

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- Choosing the larger sample solely because 412 exceeds 68

- Calling the website poll random because many different visitors could answer
- Saying only that the polls disagree without analyzing who chose to respond
- Keeping the poll open to everyone but changing only its wording

[STAR] THE PROBLEM — annotated

Suppose a school website posts the question, “Should the cafeteria go cashless?” Any visitor may answer. Of 412 responses, 71% say Yes. That same week, a teacher randomly selects three homerooms and asks every student in those rooms. Of 68 responses, 44% say Yes.

The website headline is “Students back a cashless cafeteria.” The teacher’s headline is “Students are split on going cashless.”

Two polls of student opinion

Poll	n	Yes	Selection
Website	412	71%	Visitors opt in
Homerooms	68	44%	3 rooms random

FIRST TAKE — one sentence before you work the parts

Which headline would you trust more for what students at this school think? Name one detail behind your choice.

Key: Not graded. Expect many students to choose 71 percent because 412 is larger.

[BOOK] TRUST STARTS WITH SELECTION (keep on the page)

The **population** is the whole group we want to learn about; the **sample** is the group we collect data from. Ask how the data were collected, who was included, and who was left out. A representative sample reflects the population. Chance selection helps avoid favoring one part of the population. More responses do not by themselves make a sample representative.

DOK 1 (a) Who is the population the headlines are trying to describe?

Key: All students at this school.

DOK 2 (b) In two sentences, explain how students entered each poll and which students the website poll could miss.

Key: Website visitors choose whether to answer; people who do not visit or do not feel strongly may be missed. The teacher selects homerooms by chance and asks everyone in the chosen rooms; the method’s formal name is not required today.

DOK 3 (c) Which percentage is more trustworthy for this school? Explain how selection supports your choice and why the larger response count does not settle it. Suggest one change that directly improves who enters the website sample.

Key: The 44% result is more trustworthy because the homerooms were selected by chance. The website’s 412 responses do not repair self-selection: volunteers may differ from other students. Select students by chance from the full school roster, invite only those selected, and follow up with them instead of accepting an open poll. Three rooms can still differ from the school by chance; this choice does not guarantee the exact school percentage.